

Problem 15.4.3

Find the Maclaurin series and its radius of convergence.

$$\sin 2z^2$$

Solution

Problem 15.4.16. Inverse sine.

Developing $1/\sqrt{1-z^2}$ and integrating, show that

$$\sin^{-1} z = z + \left(\frac{1}{2}\right) \frac{z^3}{3} + \left(\frac{1 \cdot 3}{2 \cdot 4}\right) \frac{z^5}{5} + \left(\frac{1 \cdot 3 \cdot 5}{2 \cdot 4 \cdot 6}\right) \frac{z^7}{7} + \cdots (|z| < 1)$$

Solution

Problem 15.4.19

Find the Taylor series with center z_0 and its radius of convergence.

$$\frac{1}{1-z}, \quad z_0 = i$$

Solution

Problem 15.4.21

Find the Taylor series with center z_0 and its radius of convergence.

$$\sin z, \quad z_0 = \frac{\pi}{2}$$

Solution

Problem 15.4.23

$$\frac{1}{(z+i)^2}, \quad z_0 = i$$

Solution